Gaurang Gupta

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Professional Summary

Software Engineer with ∼2.5 years of experience at Samsung Research architecting production-grade backend infrastructure, high-throughput data pipelines, and enterprise web applications. Proven expertise in building fault-tolerant distributed services, implementing advanced concurrency controls (e.g., atomic worker queues), and designing deterministic, inspectable AI architectures. Specialized in full-stack execution with deep system-level thinking, strict data isolation, and robust end-to-end reliability.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, SQL (PostgreSQL)
Backend & Systems: FastAPI, Node.js, RESTful APIs, Concurrency Control, Row-Level Security (RLS)
Frontend & Web: React, Next.js, Angular, Tailwind CSS, Shadcn UI, State Management (Zustand/Redux)
AI Engineering: Retrieval-Augmented Generation (RAG), LLM Auditing, Vector Search, FAISS, Embeddings
Tools & Infrastructure: Google Cloud Platform (GCP), Supabase, Firebase, Git, GitHub Actions, ArgoCD, Vercel

Professional Experience

Samsung Research & Development Institute — **Bengaluru, India**
Software Engineer — *Nov 2023 – Present*

Enterprise Web Architecture & Frontend Systems (March 2026 – Present)

- Engineered client-side input masking and strict state sanitization mechanisms for critical multi-factor authentication recovery flows, preserving immediate raw string transmission to down-stream backend servers.
- Architected localized global UI components deployed across 80+ distinct geographical zones, establishing strict behavioral parity while mapping complex, multi-lingual dynamic terminology matrices.
- Decoupled and modernised legacy Angular modules during an incremental enterprise migration to React, rewriting unsupported browser detection policies to directly improve user retention benchmarks.
- Developed interactive high-fidelity frontend modules using Tailwind CSS, converting intricate Figma design tokens into production-ready, highly accessible, reusable components.

Data Engineering & Distributed Platforms (November 2023 – March 2026)

- Designed and maintained large-scale production data pipelines on Google Cloud Platform, reliably orchestrating over 30GB of daily system telemetry and event streams under continuous operational health monitoring.
- Built an incremental ingestion architecture handling 5GB+ weekly delta loads, eliminating expensive duplicate recomputations and significantly minimizing cloud infrastructure compute costs.
- Engineered robust PII protection flows containing strict field-level cryptographic encryption and automated data lifecycle (retention/deletion) schedules to guarantee full regulatory compliance.
- Schema-engineered a revenue-critical User Transactions table with advanced indexing constraints to fuel highly accurate transaction attribution frameworks across diverse corporate business units.
- Drove a 10%+ increase in attributed revenue growth by implementing robust schema validation layers, structural quality assurance guards, and proactive real-time error-alert monitors.
- Mentored 4 software engineering interns on scalable backend design patterns and served as a core technical interviewer evaluating Data Structures, Algorithms, and Core Systems concepts.

Production Systems & AI Projects

AI Resume Analyzer — Production AI Platform — Backend — Frontend
Next.js, FastAPI, PostgreSQL, Supabase, Row-Level Security — *Personal System*

- Architected an asynchronous task worker queue via PostgreSQL row locking (`SELECT ... FOR UPDATE SKIP LOCKED`) to completely prevent worker race conditions and enable atomic task distribution.
- Built crash-resilient background workers with exponential backoff retry cycles, dead-letter state tracking, and a self-healing engine capable of recovering stale or abruptly terminated jobs.
- Enforced zero-leak data multi-tenancy by configuring PostgreSQL Row-Level Security (RLS) policies coupled with real-time analysis status updates using Supabase event streams.
- Integrated strict JSON schema evaluation for LLM outputs, alongside custom token accounting modules, prompt version control logs, and explicit latency metrics for comprehensive observability.

Self-Auditing Retrieval-Augmented Generation (RAG) System — GitHub
Python, LangChain, Gemini API, ChromaDB, Analytics Instrumentation — *Personal System*

- Designed a multi-stage, abstention-first RAG architecture that eliminates hallucination loops by programmatically choosing to reply, abstain, or prompt for clarification using deterministic verification layers.

- Implemented a pre-retrieval Query Auditor using strict specificity scoring models and token density heuristics to terminate underspecified requests prior to downstream processing.
- Engineered a post-retrieval validation service evaluating statistical distribution curves (max, mean, variance) of vector similarity scores to dynamically block generation when context evidence lacks sufficient confidence.

Latency-Governed ML Inference Service with Circuit Breakers GitHub
Python, FastAPI, Advanced Concurrency Patterns, Load Testing Proxy *Personal System*

- Developed a high-performance model routing proxy that enforces strict system latency SLOs through rolling-window P99 metrics tracking and automated admission control boundaries.
- Built resilient circuit-breaker topologies to prevent downstream model degradation cascading effects, seamlessly re-routing failing traffic to fallback models based on live telemetry flags.

Education & Certifications

National Institute of Technology (NIT), Jalandhar **Jalandhar, India**
Bachelor of Technology (B.Tech) in Electronics and Communication Engineering *Graduation: 2023*
Capstone: Embedded Adaptive Lighting Networks (Arduino) — **Certifications**: Deep Learning Specialization